PRINCIPLES OF DATA VISUALIZATION

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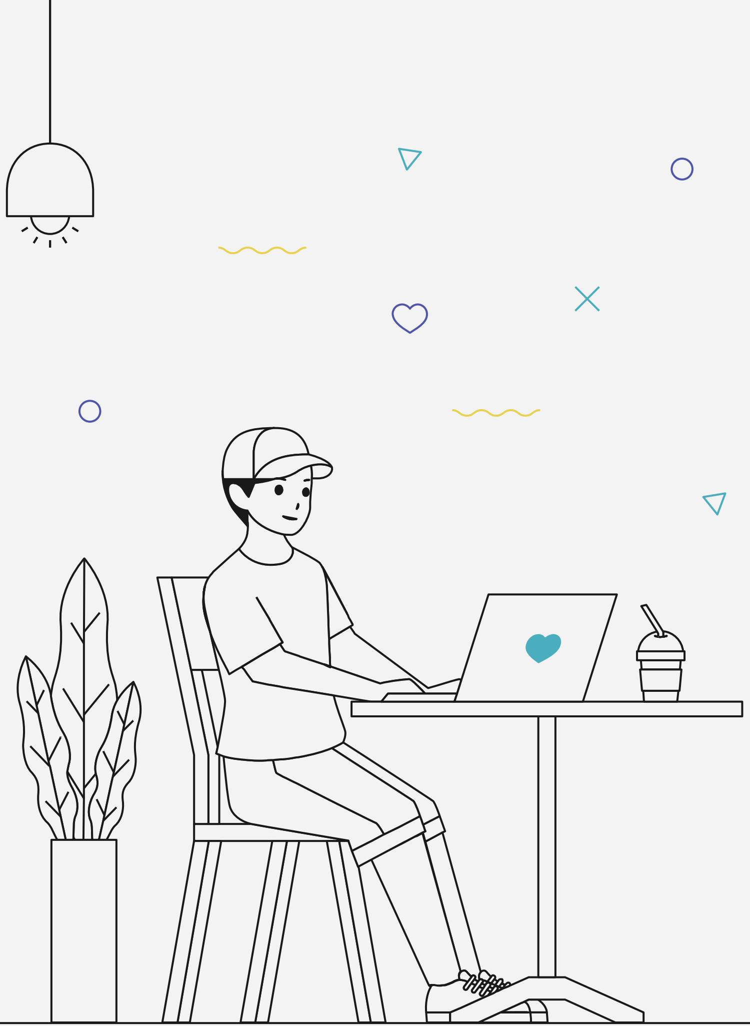
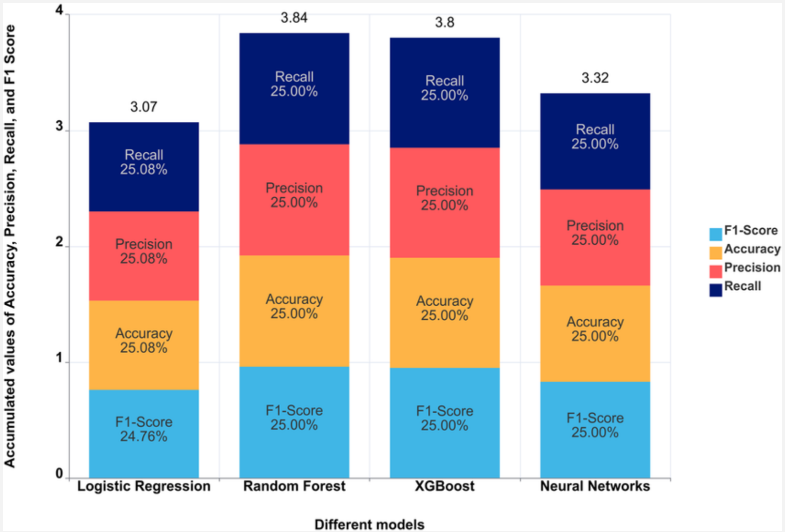
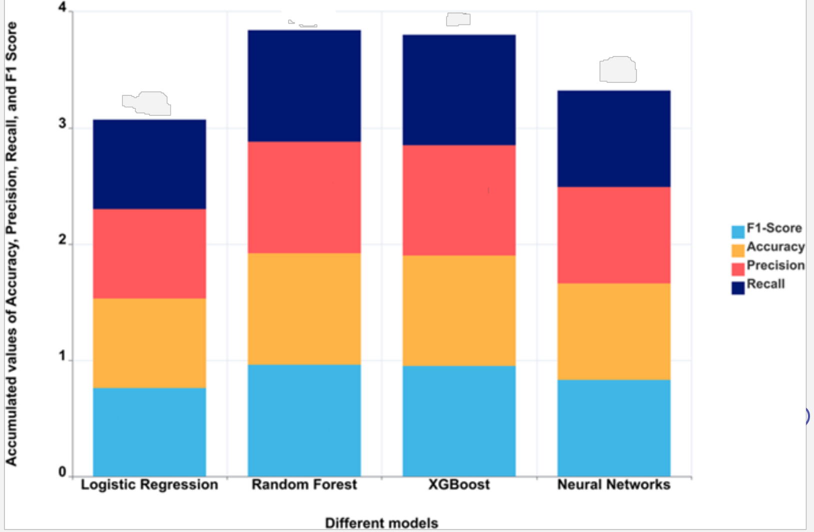
01

VISUAL PROCESSING AND PERCEPTUAL RANKINGS



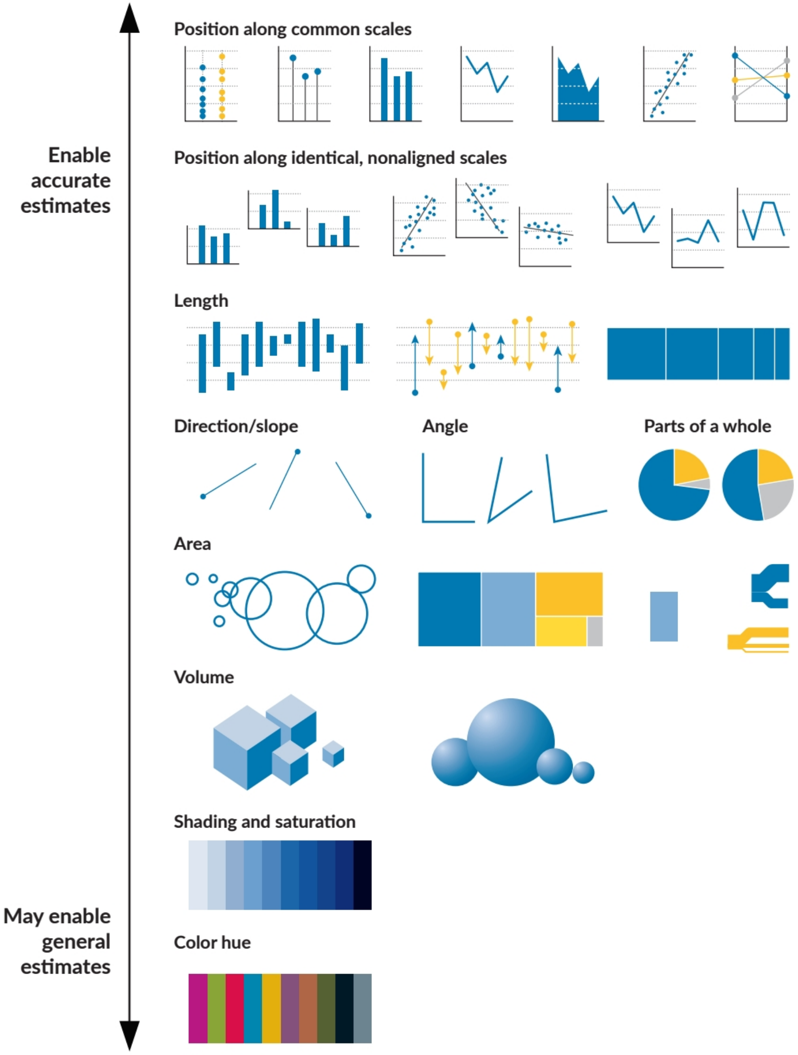




how accurately the reader can perceive the data values

Perceptual ranking diagram
- Alberto Cairo -

- ➢ Position along a common scale
- ➢ Position along non-aligned scales
- ➢ Length, direction, angle
- ➢ Area
- ➢ Volume and curvature
- ➢ Shading, and color saturation



Perceptual ranking diagram

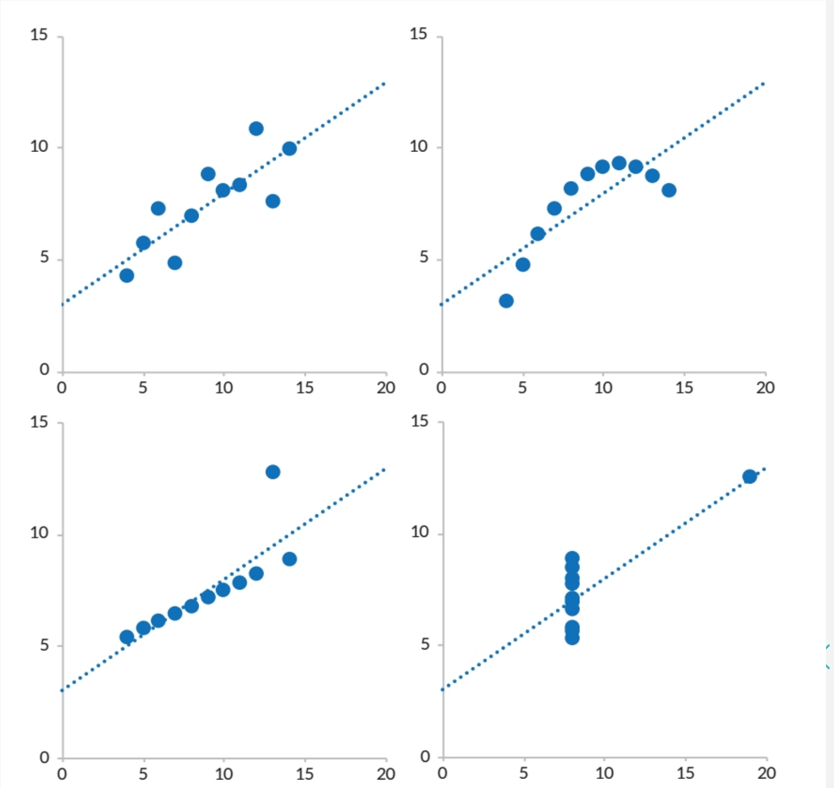
- Alberto Cairo -

Visual Encoding Type	Description	Primary Perceptual Problem	Implications for Interpretation
Position Along a Common Scale	Data points share a common axis or baseline (e.g., standard bar and line charts).	Distortion occurs when axes are truncated, unevenly scaled, or manipulated. Overplotting may reduce clarity.	Highly accurate when properly designed; vulnerable to misleading scale manipulation.
Position Along Non-Aligned Scales	Data positioned on separate or dual axes without a shared baseline.	Lack of a common reference impairs direct comparison and increases cognitive load.	May create illusory correlations and reduce comparative precision.
Length	Magnitude encoded by the length of bars or lines.	Non-zero baselines exaggerate or minimize differences; comparisons across inconsistent baselines reduce accuracy.	Moderately reliable but less precise than aligned positional encoding.
Direction, Angle, Parts of a Whole	Magnitude encoded through angles or proportional segments (e.g., pie charts).	Humans have limited ability to accurately estimate angular differences, especially among similar values.	Precision declines as the number of categories increases.
Area	Magnitude represented by two-dimensional size (e.g., bubble charts).	Area perception is nonlinear; viewers often misjudge proportional differences.	Frequently leads to over- or underestimation of relative magnitude.
Volume and Curvature	Three-dimensional representations incorporating depth and perspective.	Perspective distortion, occlusion, and foreshortening obscure accurate comparison.	Reduces clarity and perceptual accuracy; often adds aesthetic complexity without informational benefit.
Shading and Color Saturation	Magnitude encoded via differences in hue or intensity (e.g., heatmaps, choropleths).	Color discrimination is imprecise and context-dependent; accessibility limitations (e.g., color vision deficiency).	Suitable for detecting general patterns, not precise quantitative comparison.

ANSCOMBE'S QUARTET

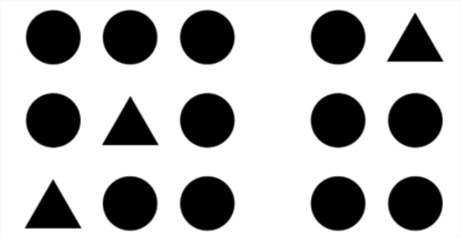
Data set		1	1	2	2	3	3	4	4
Variable		x	y	x	y	x	y	x	y
Obs. No.	1 :	10	8.0	10	9.1	10	7.5	8	6.6
	2 :	8	7.0	8	8.1	8	6.8	8	5.8
	3 :	13	7.6	13	8.7	13	12.7	8	7.7
	4 :	9	8.8	9	8.8	9	7.1	8	8.8
	5 :	11	8.3	11	9.3	11	7.8	8	8.5
	6 :	14	10.0	14	8.1	14	8.8	8	7.0
	7 :	6	7.2	6	6.1	6	6.1	8	5.3
	8 :	4	4.3	4	3.1	4	5.4	19	12.5
	9 :	12	10.8	12	9.1	12	8.2	8	5.6
	10 :	7	4.8	7	7.3	7	6.4	8	7.9
	11 :	5	5.7	5	4.7	5	5.7	8	6.9
Mean		9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
Variance		11.0	4.1	11.0	4.1	11.0	4.1	11.0	4.1
Correlation		0.816		0.816		0.816		0.817	
Regression line		y = 3 + 0.5x		y = 3 + 0.5x		y = 3 + 0.5x		y = 3 + 0.5x	

Source: Francis Anscombe

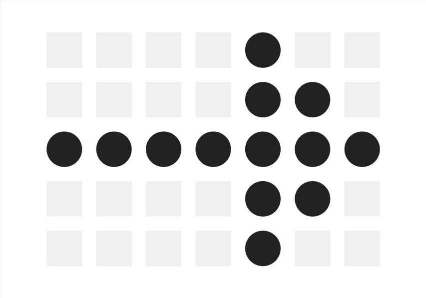


GESTALT PRINCIPLES OF VISUAL PERCEPTION

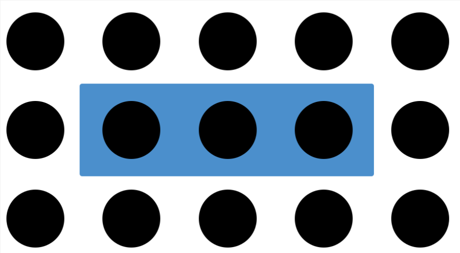
- **Proximity**



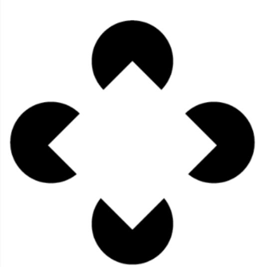
- **Similarity**



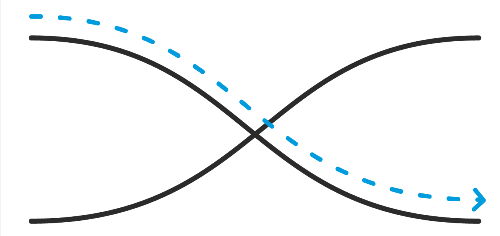
- **Enclosure**



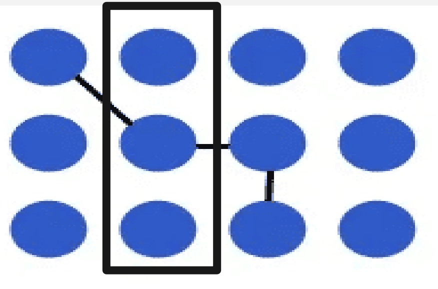
- **Closure**



- **Continuity**



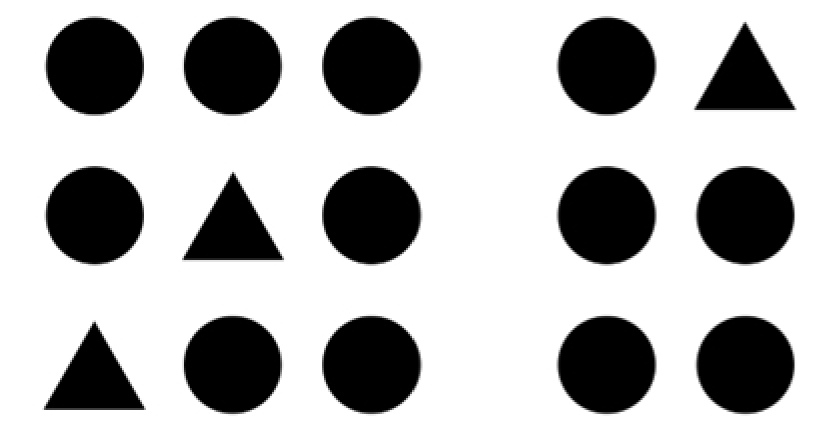
- **Connection**



Gestalt principles of visual perception

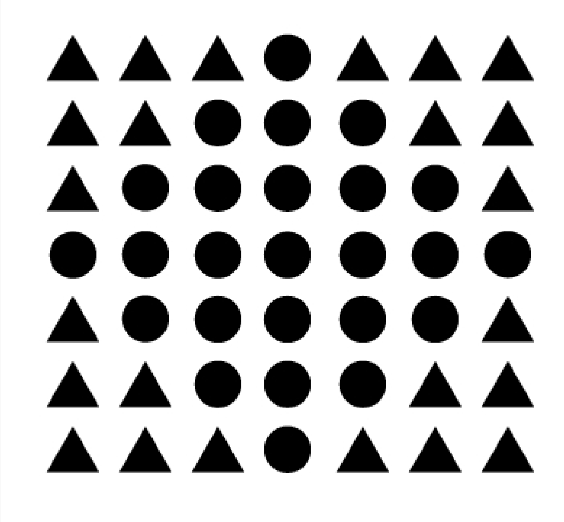
❖ **Proximity**

Objects that are close together are perceived as a group



❖ **Similarity**

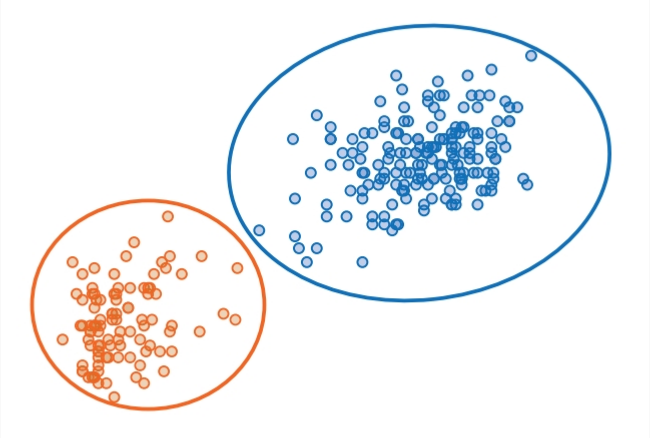
Similar objects are perceived as a group



Gestalt principles of visual perception

❖ Enclosure

Objects placed within the same enclosed boundary are perceived as belonging together.



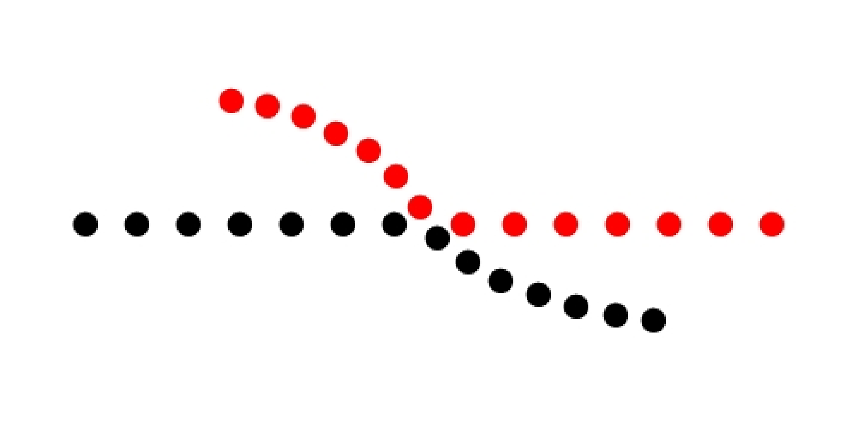
❖ Closure

The closure concept says that people like things to be simple and to fit in the constructs that are already in our heads

Gestalt principles of visual perception

- **Continuity**

Elements on a line or curve are related



- **Connection**

Elements physically linked by lines are perceived as a group more strongly than those sharing color, size, or shape

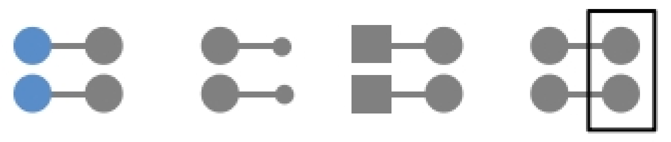
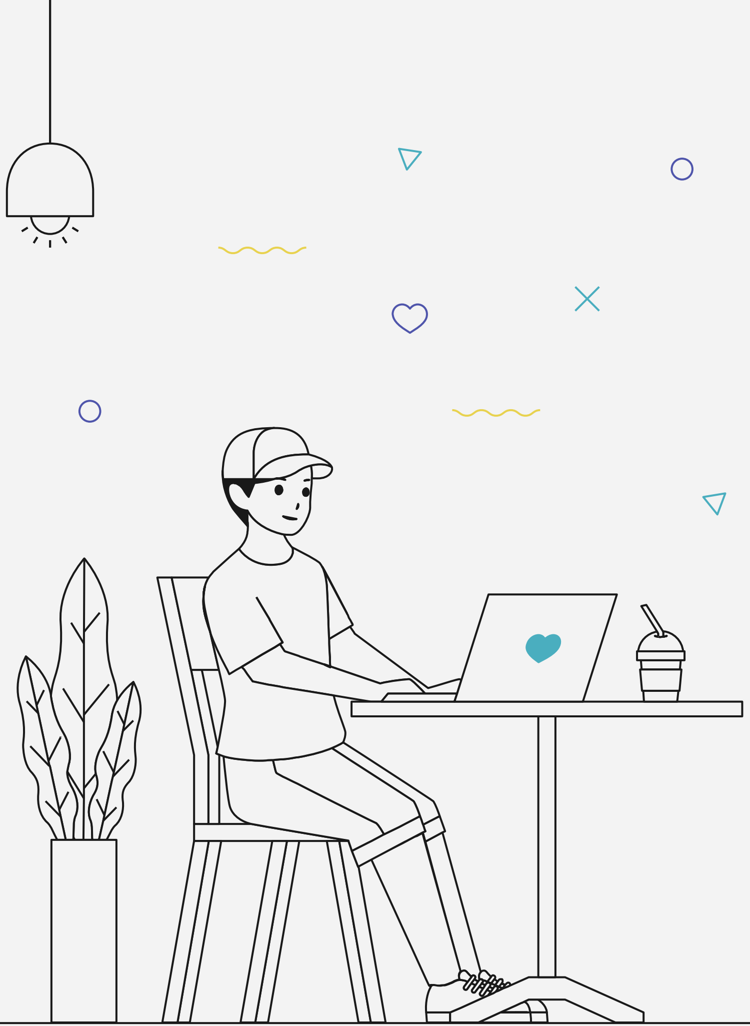


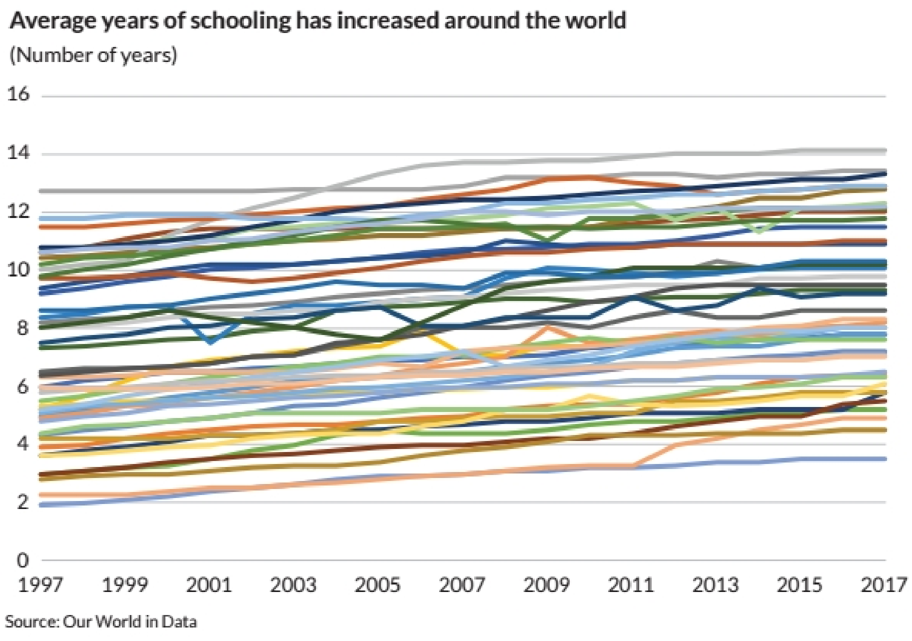
FIGURE 3.11 Gestalt principle of connection

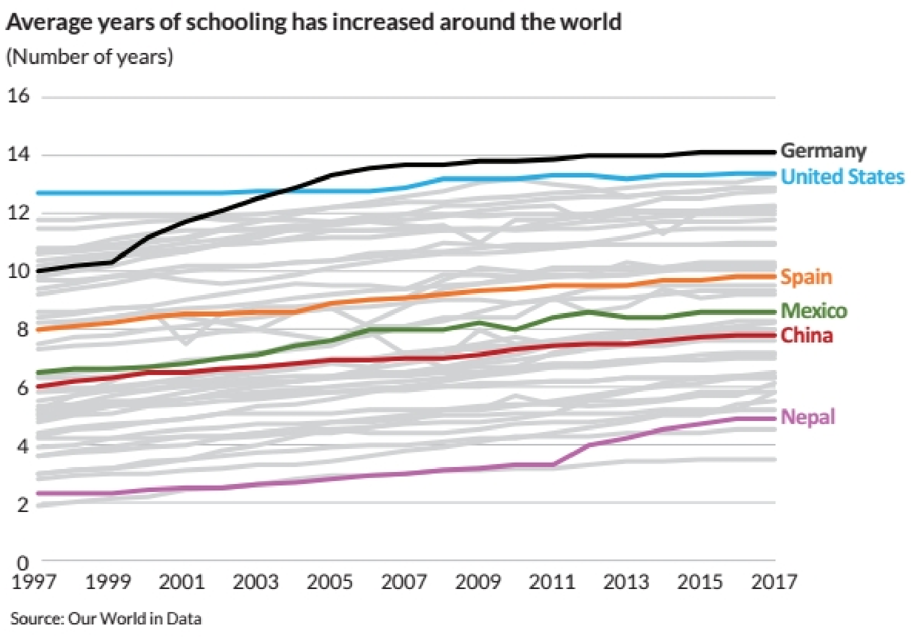
02
GUIDELINES FOR BETTER DATA VISUALIZATIONS



GUIDELINE 1: SHOW THE DATA

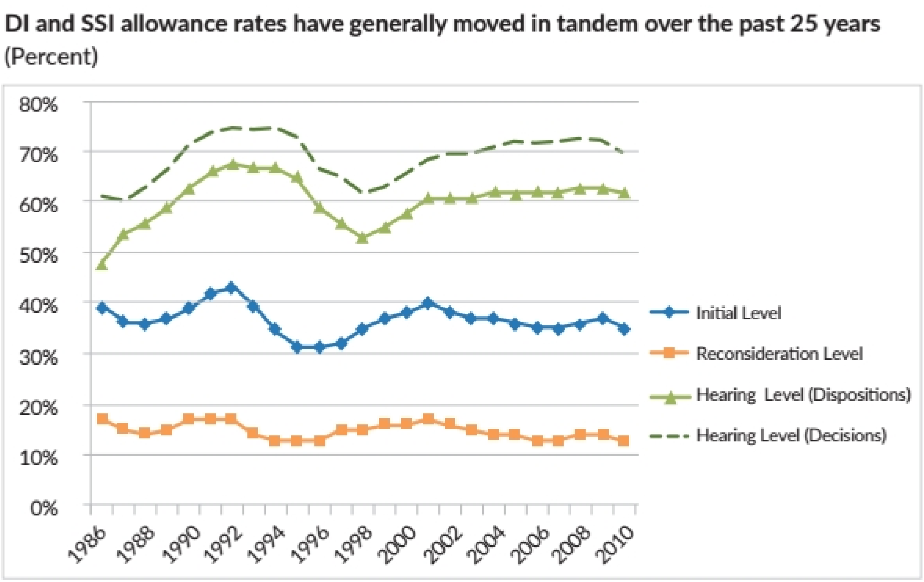
How much data to show and the best way to show it.

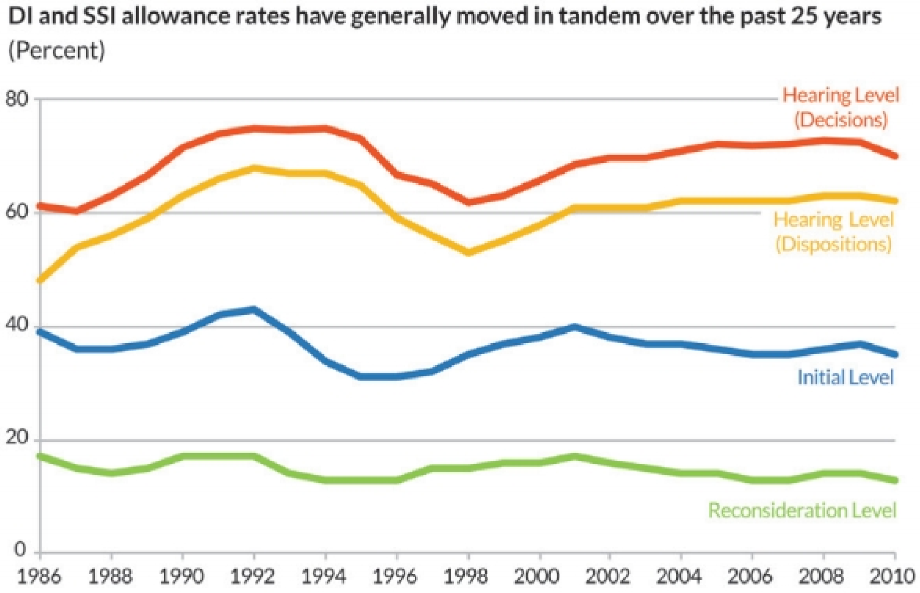




GUIDELINE 2: REDUCE THE CLUTTER

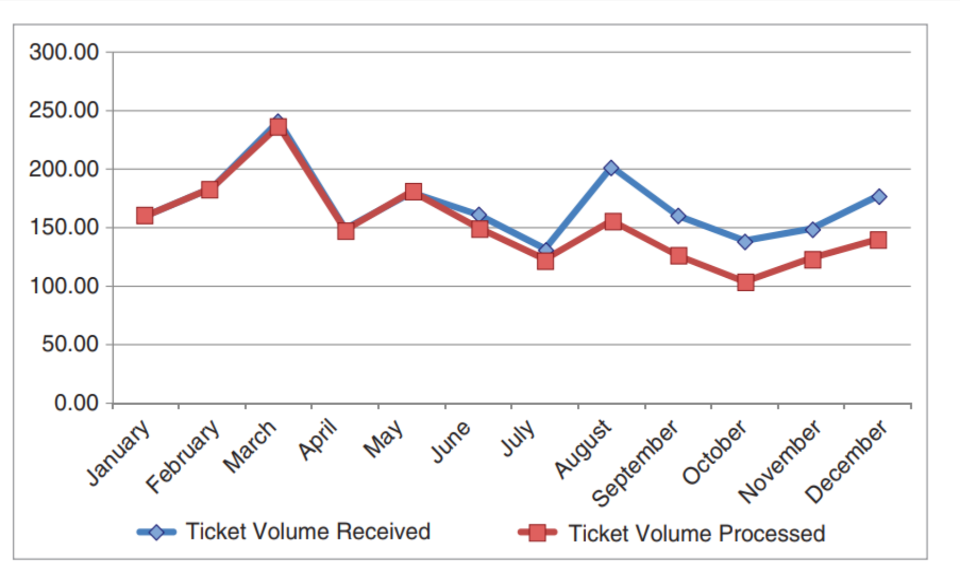
REMOVE LEGENDS WHEN POSSIBLE AND LABEL DATA DIRECTLY

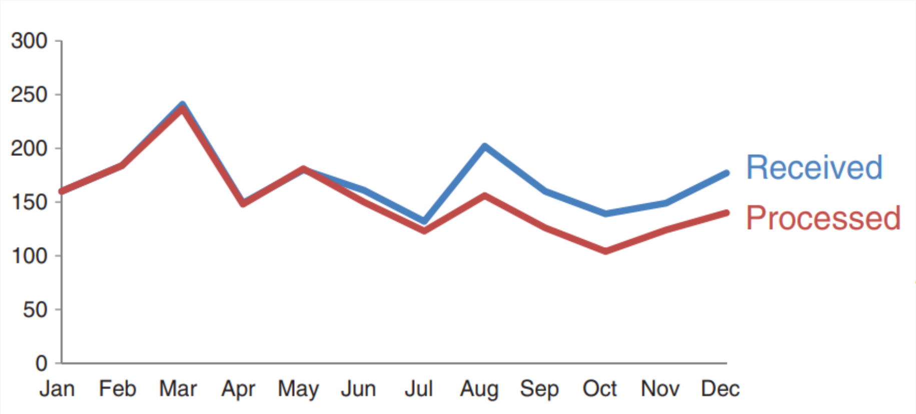




Notice the mismatch between the line order and the legend on the left chart. The redesigned version removes clutter and labels the lines directly for clarity.

GUIDELINE 2: REDUCE THE CLUTTER



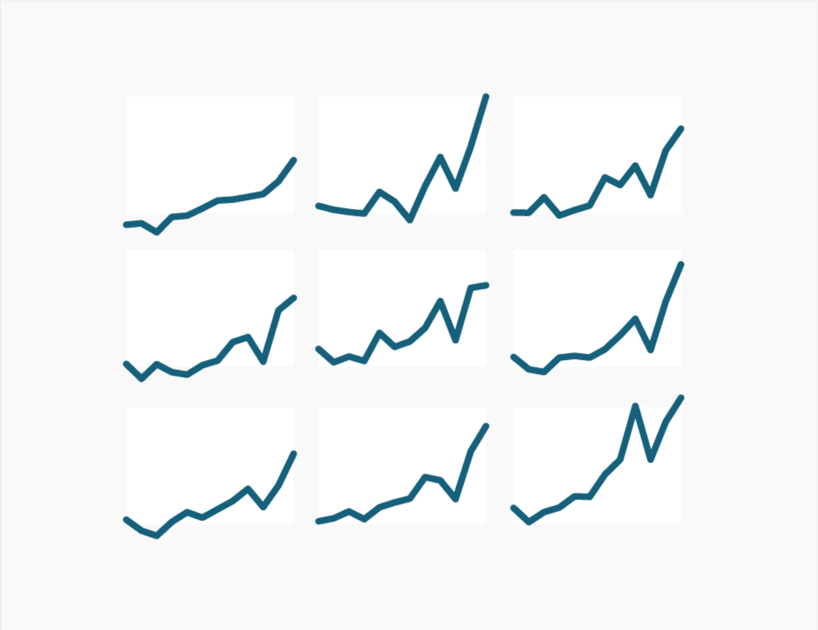


Before

After

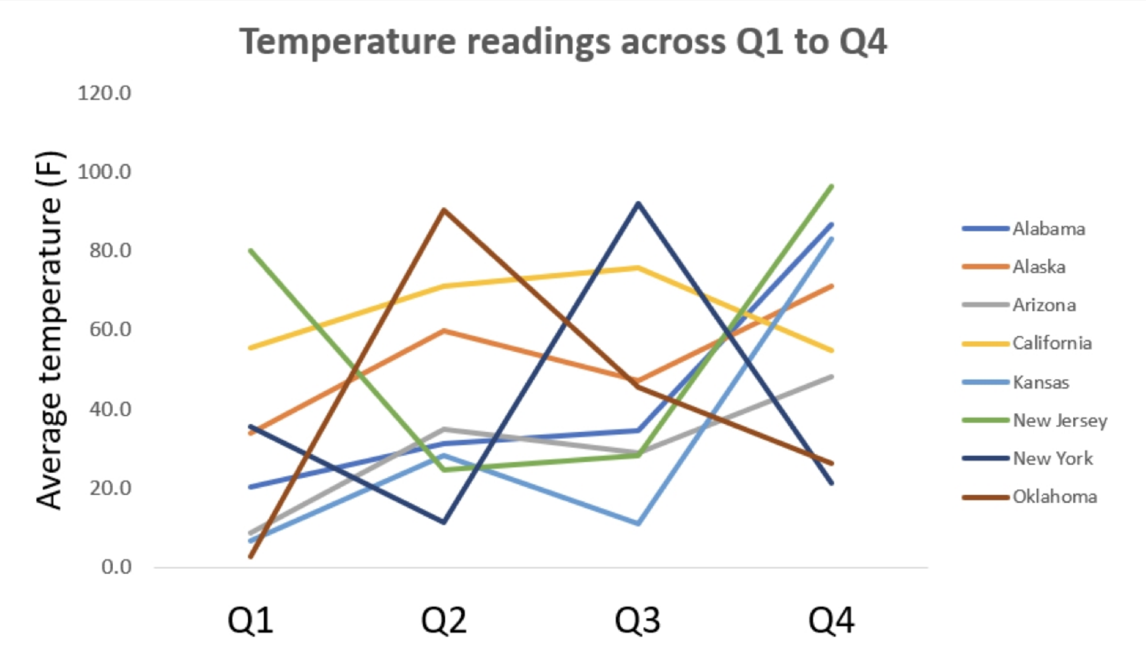
GUIDELINE 3: AVOID THE SPAGHETTI CHART

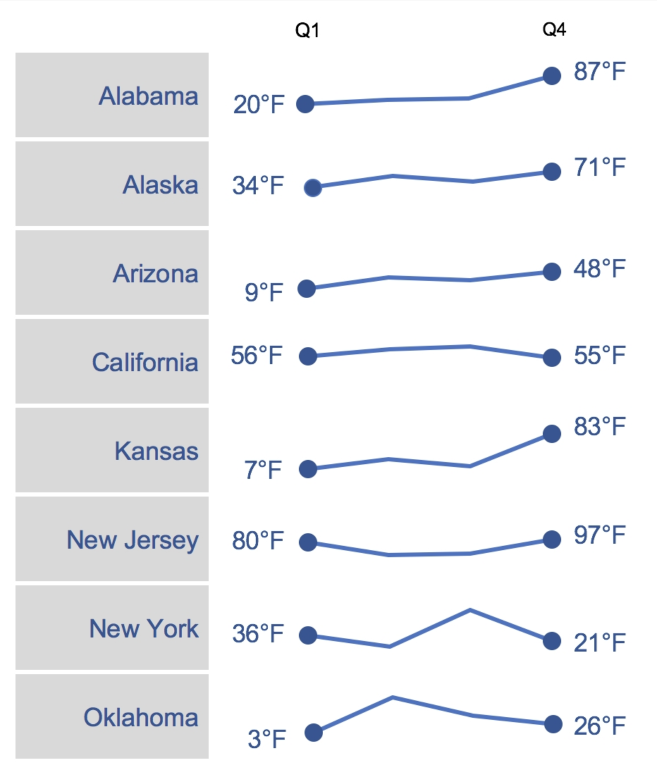
NOT IDEAL

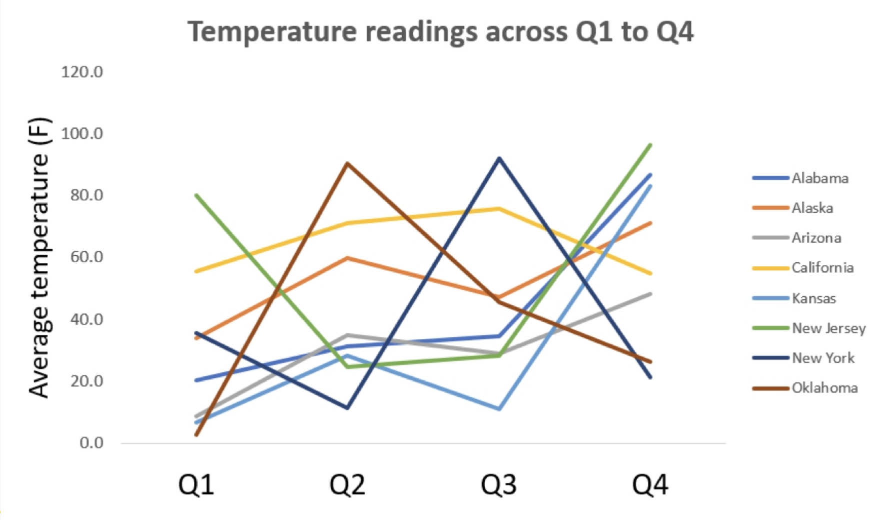


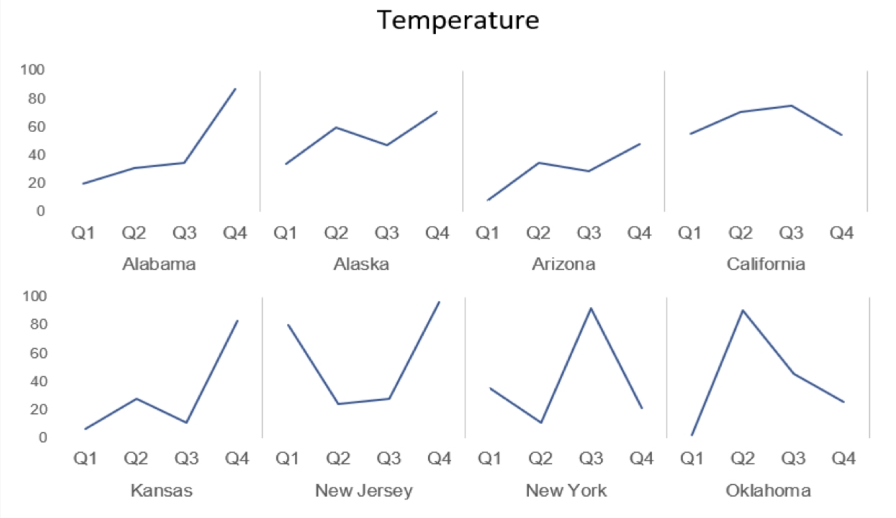
BETTER

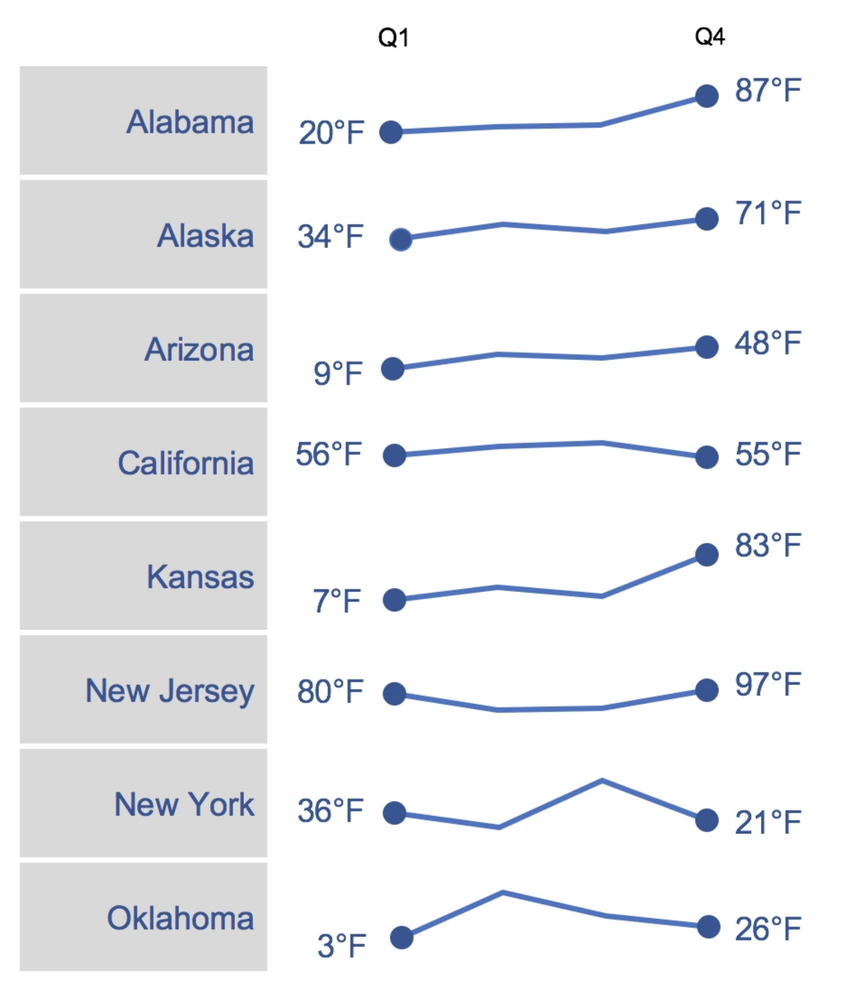
GUIDELINE 3: AVOID THE SPAGHETTI CHART



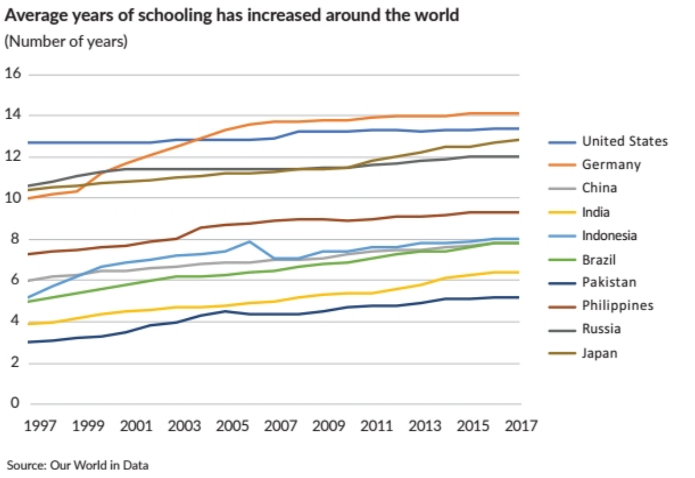


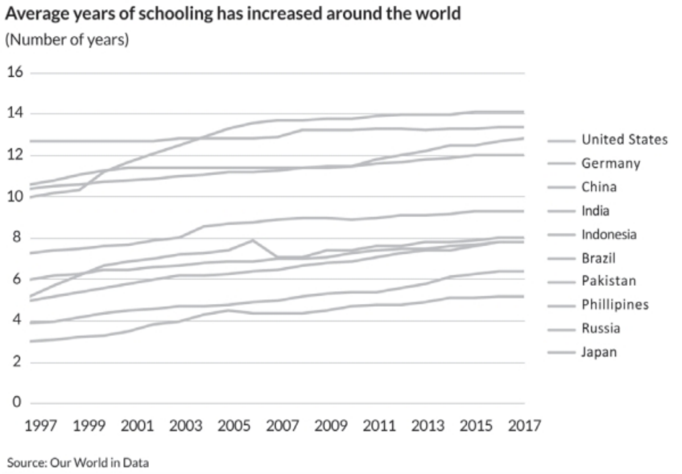

Temperature readings across Q1 to Q4
Average temperature (F)
120.0
100.0
80.0
60.0
40.0
20.0
0.0
Q1
Q2
Q3
Q4
Alabama
Alaska
Arizona
California
Kansas
New Jersey
New York
Oklahoma

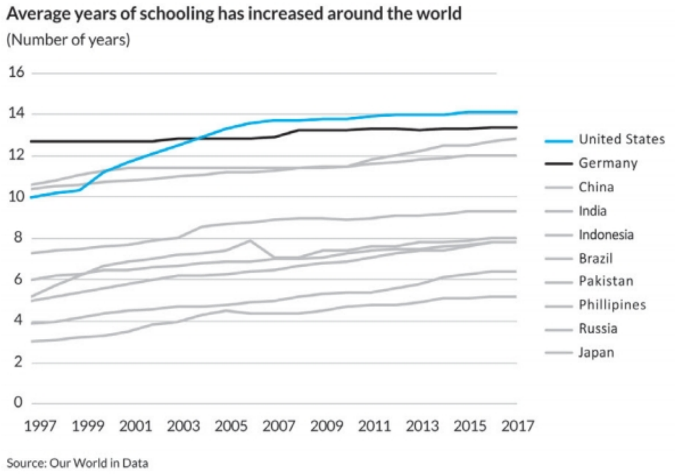

Temperature
100
80
60
40
20
0
Q1 Q2 Q3 Q4
Alabama
Alaska
Arizona
California
Kansas
New Jersey
New York
Oklahoma

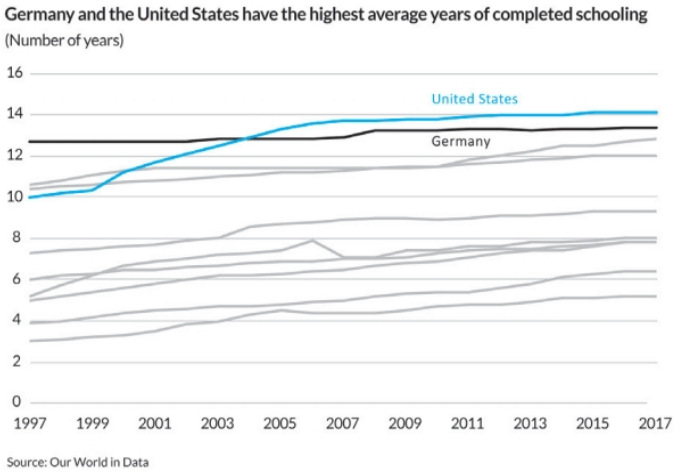

Q1
Q4
Alabama 20°F 87°F
Alaska 34°F 71°F
Arizona 9°F 48°F
California 56°F 55°F
Kansas 7°F 83°F
New Jersey 80°F 97°F
New York 36°F 21°F
Oklahoma 3°F 26°F

GUIDELINE 4: START WITH GRAY









Don’t overcomplicate

If it’s hard to read, it’s hard to do

- **Make it legible** : use a consistent, easy-to-read font
- **Keep it clean** : make your data visualization approachable by leveraging visual affordances.
- **Use straightforward language** : choose simple language over complex, choose fewer words over more words, define any specialized language with which your audience may not be familiar, and spell out acronyms (at minimum, the first time you use them or in a footnote).
- **Remove unnecessary complexity** : when making a choice between simple and complicated, favor simple

Thanks!

Do you have any questions?

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